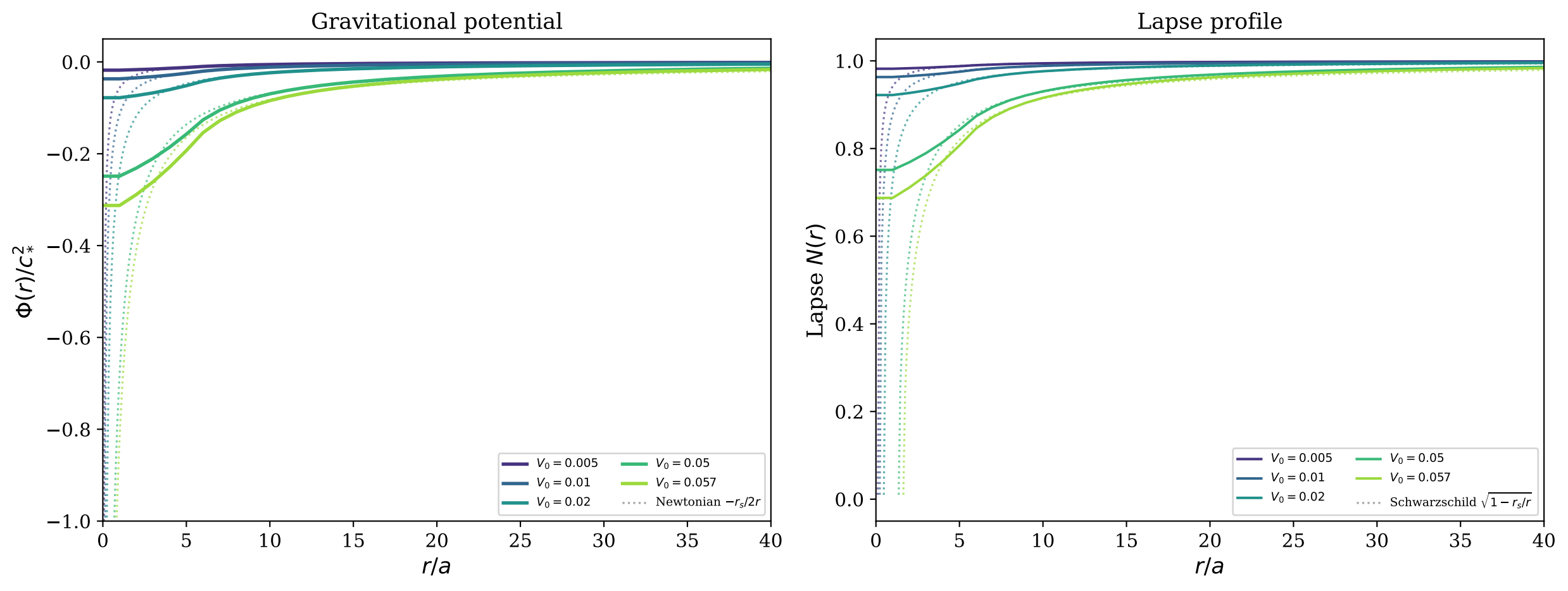
Gravitational potential
$\Phi(r)/c_*^2$
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
5
10
15
20
25
30
35
40
r/a
$V_0 = 0.005$
$V_0 = 0.01$
$V_0 = 0.02$
$V_0 = 0.05$
$V_0 = 0.057$
Newtonian $-r_s/2r$
Lapse profile
Lapse $N(r)$
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
30
35
40
r/a
$V_0 = 0.005$
$V_0 = 0.01$
$V_0 = 0.02$
$V_0 = 0.05$
$V_0 = 0.057$
Schwarzschild $\sqrt{1 - r_s/r}$